- Functional Requirements
1. User submits long URL and gets short URL
2. System generates a short URL
3. When someone clicks the short URL, it redirects to the original URL.

- Non Functional Requirements
1. High availability
2. Low latency
3. Highly scalable
4. Fault tolerant

- High Level Architecture

Client
|
API Gateway
|
URL Shortening Service
|
Database
|
Cache (Redis)

- Database Design

MySQL
Primary key → id

Table:

Id	Long URL	Short URL	created_at	

- Hashing
  Hash the URL
  hash (URL) → short URL

- Random ID
  generate random ID

- Caching
  To speed up redirection, use cache like Redis

- Scaling the System
  1. Load Balancer like NGINX
  2. Scale databases using the techniques such as Replication and Sharding.

Final Architecture

User
|
DNS
|
Load Balancer
|
Application Server
|
Cache (Redis) — Database